



REPUBLIC OF YEMEN  
SANA'A UNIVERSITY  
FACULTY OF ENGINEERING  
MECHATRONICS ENGINEERING DEPARTMENT



## **A non-contact\_AC\_line\_checker.**

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System(Parallel.)

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Contact.

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## Introduction

Electric current is a flow of electrical charges run over connector and to know this charges exist or not in any conductor or not without any risks you need a device(tool).

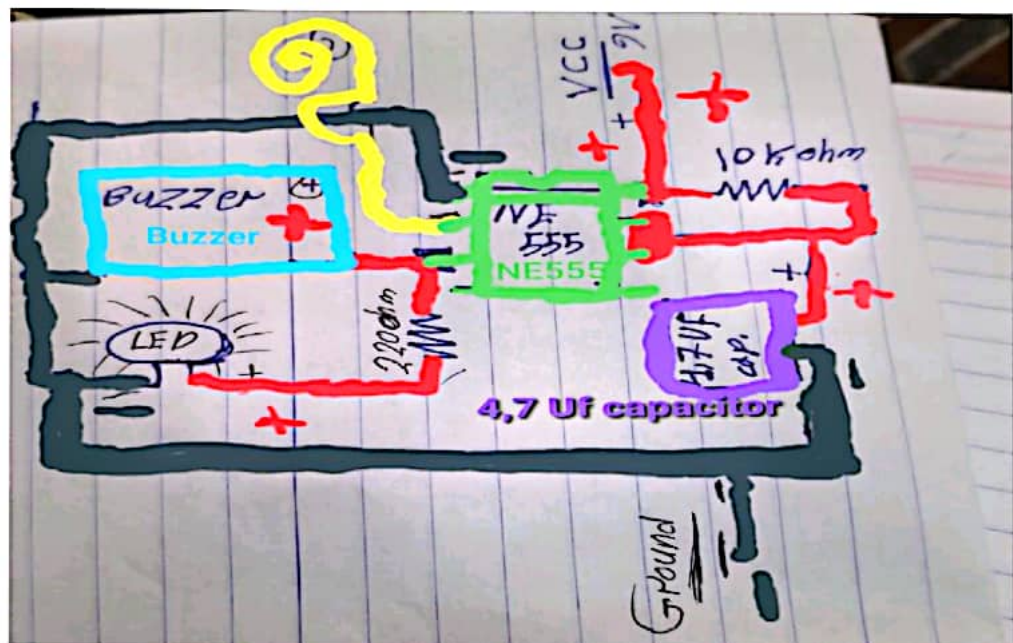
This device works without the need to connect with this conductor and this is an advantage and a matter of safety concerns for customer.

These things will work the success of the project on a wide cast.

## procedures

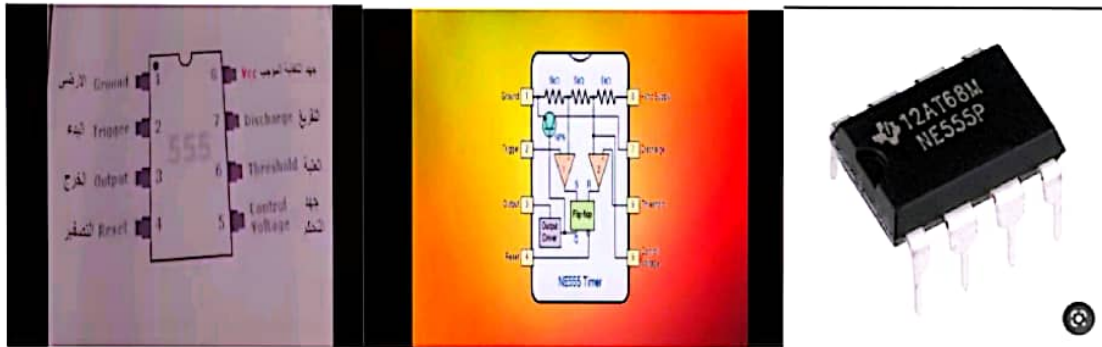
### 1. Tools

#### a) Software:



## b) Hardware:

- NE555



- 10k  $\Omega$  resistor
- buzzer
- 220 $\Omega$  resistor
- Led
- 4.7, uf, capacitor
- voltage source 9V
- copper connector
- welding tool

**2.designe process:** is connected vcc of the piece NE555 anode of source and connecting the ground via negative pole of the source

The entrance 67of the piece NE555 is connected with a type of electrical resistance  $10k\Omega$  and the other type via the positive pole of the source

The widget positive side of the buzzer is connected via entrance 3 of the NE555 and the other side with negative pole of the source

Resistance  $220\Omega$  is connecting with entrance 3 of NE555 from one side and by the photodiode from other side and connect the other side of photodiode by the negative pole of the source

Connect the positive side of the capacitor by the entrance 67 of the part NE555 and the positive side with the negative pole of the source.

Connect a coil of insulated copper wire by the entrance 2 of the piece NE555 and it is directed to the area where the electric current need to be known or not.

## Conclusion .

Good luck for everyone

## Reference .

From the channel in YouTube are called electronics.